BIR VA KO'P O'ZGARUVCHILI FUNKSIYA

1. $X = Y = R$ da $y = x^2$ funksiyaning aniqlanish sohasini toping.

Yechish. $X = Y = R$ da $y = x^2$ funksiyaning aniqlanish sohasini $D(y) = (-\infty; +\infty)$ bo'ladi. ▸

2. $f(x) = \sqrt{1-2x} + 3\arcsin\frac{3x-1}{2}$ funksiyaning aniqlanish sohasini toping.

Yechish. Birinchi qo'shiluvchi $1-2x \geq 0$ da haqiqiy qiymatlarni qabul qiladi, ikkinchi qo'shiluvchi esa $-1 \leq \frac{3x-1}{2} \leq 1$ bo'lganda. Shunday qilib, funksiyaning aniqlanish sohasini topish uchun

$$\begin{cases} 1-2x \geq 0 \\ \frac{3x-1}{2} \leq 1 \\ \frac{3x-1}{2} \geq -1 \end{cases}$$ tengsizliklar sistemasini yechamiz

$$x \leq \frac{1}{2}, \quad x \leq 1, \quad x \geq -\frac{1}{3}$$

Demak, $D(f) = \left[-\frac{1}{3}; \frac{1}{2}\right]$. ▸

Quyidagi funksiyalarning aniqlanish sohasini toping:

3. $y = \lg(x^2 - 4x + 3)$

4. $y = \arcsin(3x - 4)$

5. $X = Y = R$ da $y = x^2$ funksiyaning qiymatlar to'plamini toping.

Yechish. $X = Y = R$ da $y = x^2$ funksiyaning qiymatlar to'plami $E(y) = [0;+\infty)$ bo'ladi. ▸

6. $y = 1 + 2^{x+1}$ funksiyaning qiymatlar to'plamini toping.

Yechish. $y = 2^{x+1} = 2 \cdot 2^x$ ko'rsatkichli funksiya, uning qiymatlar to'plami $y \in (0;+\infty)$, demak berilgan funksiyaning qiymatlar to'plami $(1;+\infty)$ bo'ladi yoki berilgan funksiyaning qiymatlar to'plami uning teskari funksiyasi $x = \log_2(y-1) - 1$ ning aniqlanish sohasi $y > 1$ bilan ustma-ust tushadi, shuning uchun $E(y) = (1;+\infty)$. ▸

Quyidagi funksiyalarning qiymatlar to'plamini toping:

7. $y = x + \dfrac{1}{x}$

8. $y = \sqrt{-x^2 - x + 2}$

9. $y = 2x + 1$ funksiya $(-\infty;+\infty)$ da monotonligini ko'rsating.

Yechish. $y = 2x + 1$ funksiya $(-\infty;+\infty)$ da o'suvchi, chunki $x_1 < x_2$ bo'lsa, u holda

$f(x_2) - f(x_1) = 2x_2 + 1 - (2x_1 + 1) = 2(x_2 - x_1) > 0$ bo'ladi va $f(x_1) < f(x_2)$ tengsizlik kelib chiqadi. ▶

Quyidagi funksiyalarning har biri ko'rsatilgan oraliqlarda monoton ekanligini ko'rsating:

10. $f(x) = 2x - 1; \quad x \in (-\infty;+\infty)$,

11. $f(x) = 2 - 3x; \quad x \in (-\infty;+\infty)$,

12. $f(x)=\lg\left(x+\sqrt{x^2+1}\right)$ funksiyaning juft yoki toqligini tekshiring.

Yechish.

$$f(-x)=\lg\left(-x+\sqrt{(-x)^2+1}\right)=\lg\frac{\left(x+\sqrt{x^2+1}\right)\left(-x+\sqrt{x^2+1}\right)}{\left(x+\sqrt{x^2+1}\right)}=$$

$$=\lg\frac{1}{\left(x+\sqrt{x^2+1}\right)}=\lg\left(x+\sqrt{x^2+1}\right)^{-1}=-\lg\left(x+\sqrt{x^2+1}\right)=-f(x).$$

Demak, $f(x)=\lg\left(x+\sqrt{x^2+1}\right)$ toq funksiya ekan. ▶

Quyidagi funksiyalarning juft yoki toqligini tekshiring:

13. $y=\dfrac{1}{x+\dfrac{1}{x+\dfrac{1}{x}}}$

14. $y=x^2\cos x$

15. $y=Sin6x+tg4x$ funksiyaning asosiy davrini toping.

Yechish. Birinchi qo‘shiluvchi uchun asosiy davr $\frac{2\pi}{6}=\frac{\pi}{3}$ bo‘ladi, ikkinchi qo‘shiluvchi uchun davr $\frac{\pi}{4}$ bo‘ladi. $\frac{\pi}{3}$ va $\frac{\pi}{4}$ sonlarining eng kichik umumiy karralisi bo‘lgan π funksiyaning asosiy davri bo‘ladi. $f(x+T)=f(x);\ T=\pi$. ▶

16. $y=\cos\left(\sqrt{x}\right)^2$ funksiyaning asosiy davrini toping.

Yechish. $y=\cos\left(\sqrt{x}\right)^2$ funksiya davriy emas, chunki uning aniqlanish sohasi bo‘lgan $D(y)=[0;+\infty)$ to‘plam quyidan chegaralanganligi uchun davriy to‘plam emas. ▶

Funksiyalarning asosiy davrini toping:

17. $y = 2$

18. $y = Sin\frac{3}{2}x + 1$

19. $f(x) = x^3 - 3x^2 + 3x$ funksiyaga teskari funksiyani toping:

Yechish. $y = (x-1)^3 + 1$, $(x-1)^3 = y - 1$ va $x = \sqrt[3]{y-1} + 1$. x ni y ga almashtirib quyidagi teskari funksiyani olamiz: $f^{-1}(x) = \sqrt[3]{x-1} + 1$.

▸

Quyidagi funksiyalarning teskari funksiyalarini quring:

20. $y = 2 - 5x$

21. $y = \frac{2}{x+1} - 3$

22. $f(x, y) = \arcsin(x + y)$ funksiyaning aniqlanish sohasini toping:

Yechish. $f(x, y) = \arcsin(x + y)$ funksiya ma'noga ega bo'lishi uchun x va y lar ushbu $-1 \le x + y \le 1$ munosabatda bo'lishi lozim. Bu tengsizliklarni tekislikning $x + y + 1 = 0$ va $x + y - 1 = 0$ to'g'ri chiziqlar orasidagi nuqtalarning koordinatalari qanoatlantiradi. Demak, $D(f) = \{(x, y) \in R^2; |x + y| \le 1\}$. ▸

Quyidagi funksiyalarning aniqlanish sohasini toping va koordinatalar tekisligida tasvirlang:

23. $f(x,y)=\sqrt{x+y}$

24. $f(x,y)=\sqrt{-x}+\sqrt{y}$

? Quyidagi funksiyalarning aniqlanish sohasini toping:

25. $y=\dfrac{1}{\lg(1-x)}+\sqrt{x+2}$

26. $y=\sqrt{Sin(x)}-\sqrt{9-x^2}$

27. $y=\arcsin\dfrac{1+x^2}{2x}$

28. $y=x^{\frac{1}{\lg x}}$

29. $y=tg\sqrt{16-x^2}$

30. $y=\dfrac{1}{\sqrt{|x|-2|x-1|}}$

31. $f(x)=\ln Cos(x)$.

32. $y=\sqrt{x^2+x-2}$

33. $y=\log_2(-x)$

34. $y=\arccos\left(\dfrac{x}{2}-1\right)$

35. $y=\sqrt{3^x-5^x}$

36. $y=\dfrac{1}{xe^x}$

37. $y=Sin\sqrt{x-3}$

? Quyidagi funksiyalarning qiymatlar to‘plamini toping:

38. $y=\dfrac{x+1}{x-2}$

39. $y=\dfrac{1}{x^2+1}+1$

40. $y=\dfrac{1}{\arcsin(1-x)}$

41. $y=Sin^2x-Cos^2x$

42. $y=-x^2-5x+6$

43. $y=1-|x|$

44. $y=\dfrac{2-x}{x+3}$

45. $y=\lg\dfrac{2}{\sqrt{4-x^2}}$

46. $y=\sqrt{Sin\,x-1}$

? Quyidagi funksiyalarning har biri ko‘rsatilgan oraliqlarda monoton ekanligini ko‘rsating:

47. $f(x)=x^2+2x+5;\quad x\in(-\infty;-1)\cup(-1;+\infty)$,

48. $f(x)=\cos x;\quad x\in[0;\pi]$,

49. $f(x)=\sin x;\quad x\in\left(-\frac{\pi}{2};\frac{\pi}{2}\right)$,

50. $f(x)=|x|-x;\quad x\in(-\infty;+\infty)$,

51. $f(x)=|x|+x;\quad x\in(-\infty;+\infty)$.

? Quyidagi funksiyalarning juft yoki toqligini tekshiring:

52. $y=|x|-5e^{x^2}$

53. $y=\frac{|Sinx|}{1-Cosx}$

54. $y=Sin(arcCos\,x)$

55. $y=\left|\frac{10^x+1}{10^x-1}\right|$

56. $y=\lg\frac{x+3}{x-3}$

57. $f(x)=\begin{cases}1, & agar\ \ x-ratsional\ \ son\ bo`lsa\\ -1, & agar\ \ x-irratsional\ \ son\ bo`lsa\end{cases}$

58. $y=x^2Sin\frac{1}{x}$

59. $y=\lg Cos\,x$

60. $y=\frac{16^x-1}{4^x}$

61. $y=x^4Sin7x$

? Funksiyalarning asosiy davrini toping:

62. $y=Sin\,x-Cos\,x$

63. $y=Sin2x-2tg\left(\frac{x}{2}\right)$

64. $y=Cos^2x$

65. $y=x-[x]$

66. $y=-2Cos\frac{x}{3}+1$

67. $y=Sin\frac{x}{3}+ctg\frac{x}{4}$

68. $y=2^{Sin2x}\cdot 2^{Cos2x}$

69. $y=\log_2 Sin\ x$

? Quyidagi funksiyalarning teskari funksiyalarini quring:

70. $y=\dfrac{x+2}{3-2x}$

71. $y=\dfrac{1}{2-x}+4$

Quyidagi funksiyalarning aniqlanish sohasini toping va koordinatalar tekisligida tasvirlang:

72. $f(x,y)=\arccos\dfrac{x^2+y^2}{9}$

73. $f(x,y)=\sqrt{x-\sqrt{y}}$

74. $f(x,y)=1+\sqrt{-(x-y)}^2$

75. $f(x,y)=\dfrac{1}{x-1}+\dfrac{1}{y}$

76. $f(x,y)=\dfrac{\sqrt{4x-y^2}}{\lg(1-x^2-y^2)}$

77. $f(x,y)=\sqrt{x^2-4}+\sqrt{1-y^2}$

78. $f(x,y)=\sqrt{x}+\sqrt{y}$

79. $f(x,y)=\sqrt{1-x^2-y^2}$

80. $f(x,y)=\sqrt{y\,Sin\,x}$

81. $f(x,y)=\ln(x+y)$

82. $f(x,y)=\ln\left(\dfrac{x^2}{9}-\dfrac{y^2}{4}-1\right)$

83. $f(x,y)=\sqrt{Cos(x^2+y^2)}$

84. $f(x,y)=\sqrt{x+y}-\sqrt{x-y}$

85. $f(x,y)=\sqrt{25-x^2-y^2}$

86. $f(x,y)=\ln(y^2-4x+8)$

87. $f(x,y)=\dfrac{1}{2-x^2-y^2}$

88. $f(x,y,z)=\dfrac{z}{x-y}$

89. $f(x,y)=\arcsin\dfrac{y-1}{x}$

90. $f(x,y)=\arcsin\dfrac{x}{2}+\arccos\dfrac{y}{2}$

91. $f(x,y)=\dfrac{1}{\sqrt{x}}+\dfrac{1}{\sqrt{y}}+\dfrac{1}{\sqrt{z}}$